



200L 2000L fermenters BAILUN BIO-200SJA 2000SJA--GMP



BAILUNBIO-50 500 5000SJ---GMP



OKL GMP fermenter

QUOTATION

SEQ	ITEM DESCRIPTION	QTY/SET	TYPE	Unit Price (USD\$)	Amount (USD\$)
1	7000L GMP automatic stainless steel fermenter	1	BAILUNBIO-7000SJA-GMP	230,000	230,000
2	Documents FAT, SAT, IQ, OQ PQ	1		8,000	8,000
3	Installation, debugging, training charge: USD\$300/day/person Customer affords transportation from China to customer site and accommodation at local place.				

Remarks:

1. The prices are FOB Shanghai.
2. The price mentioned above is included Chinese internal transportation expenses, installation and debugging expenses and the personnel training expenses. Overseas Customer need pay transportation, accommodation and flight charge for one or two people
3. Payment item: 50%TT as deposit, 50%TT before FAT and shipment.

4. Delivery date: 90-120th working day after signing the contract.
5. Guarantee: 2 years. We will solve the quality problem by the ways as internet training, changing parts and engineer arriving customer site for repairing for free.

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Description for 7000L fermentor system

1、Equipment description	
Vessel	TTL volume: 7000L
	D:H rate: 1: 2.5;
	Working volume: 70%-80%
	The vessel material is stainless steel 316L. The internal surface of vessel is polished with Ra ≤0.4um
	The jacket is 304, the surface polishing precision Ra≤1.2μm, all welds are firm and neat and beautiful. The material for automatic welding shall comply with the chemical composition described in the DT3 table of ASME BPE2009
	500L vessel thickness: 6mm.
	Vessel design pressure: 0.1-0.35Mpa
	Jacket design pressure: 0.4 Mpa, Design temperature: 150 °C
	Provide pressure vessel certificate, with insulation layer
	Vessel function: with special discharge valve for fermentation liquid discharge and transferring. Adopt large sight glass for liquid level observation, 12V safety sight light.
	Vessel interfaces for intake sterilization filter, exhaust condenser, air intake system with air distributor, feed port, cleaning port, discharge port, sampling port, 4 replenishment material ports, pressure gauge port.
Stir system	Stir method: upper single mechanical seal. Single axis, micro taper and high speed type. Material contacting with the broth is 316L. Precision processing, ideal dynamic balance performance, good rigidity, long-term use without deformation; The entire system is safe and stable and long life.
	Stirring motor: adopts SEW AC motor. Variable frequency, low noise (below 60dB), high temperature resistance (above 125 °C);
	Stirring type: flat vertical paddle / rushton impeller (2 grades), middle-grade is pitched paddle, mechanical antifoaming paddle (1 grade), four baffles;
	Stirring speed: 50 ~ 1000rpm stepless speed change; accuracy: ± 1rpm;
	Stirring power: 1500L 5.5KW
Sterilization system	Automatic sterilization for empty tank and tank filled liquid, and supports manual operation sterilization; The pure steam main pipe is provided with a steam-water separation device, a pressure reduction device and a filtration device. The pure steam enters the tank through the air filter element; The steam pipeline is added with a steam trap, and steam enters the jacket ;

	Sterilization temperature requirements: 121 ~ 125 °C. From the start of sterilization to cooling to 30 °C, the sterilization for empty tank does not exceed 1h, the sterilization for tank filled with liquid does not exceed 1.5h
Sampling system	Adopt aseptic sampling valve can realize multiple aseptic sampling in situ, without dead angles and material accumulation, and can be independently sterilized in situ. The sampling position is set at the bottom of the tank, and it is connected to the transplanting pipeline. The sampling valve uses a sampling valve and a separate pure steam sterilization pipeline system. The sampling valve is a sanitary-grade closed sterile sampling valve, which can be used separately, multiple times Sterilized in situ.
N ₂ intake system	The filter element interface is a standard 5-inch, 0.22 micron air filter element. The ventilation mode is deep ventilation. The deep ventilation pipeline enters from the lower side shell of the tank body and the pipeline has a non-return device. The circular air distributor can evenly ventilate around; The N₂ intake method is automatic and manual. The mass flow meter is automatically controlled, and the rotor flow meter is manually assisted. On-line integrity detection interface is reserved for intake line. The mass flow meter range: : 0-1VVM.
Operating mode	The deep ventilation sterilization mode is set according to the sterilization temperature, and related the air valve, steam valve and exhaust valve; The pressure-holding mode is set according to the tank pressure and tank temperature, and related the intake valve, exhaust valve, cooling water valve, etc.
Gas filtration	The off-gas exhausts through the condensing section, the gas sterilization filter and the pressure regulating valve in order, and then is discharged. The filter element interface is connected to a standard 5-inch, 0.22 micron air filter element, and the filter can be SIP; The device for anti-running fluid and filter element blocking are reliable, and the exhaust gas filter is equipped with a heating device. On-line integrity detection interface is reserved
Pressure control	German BD pressure electrode is for digital display. Automatically control the required tank pressure through off gas adjustment, etc., and record the pressure change curve during the fermentation process. Range: -0.1~0.25Map; precision: ± 0.01
PH control	Using the Hamilton intelligent pH electrode for fermentation, Control Range: 2~12, display range: 0~14, control precision: ±0.02, resolution: 0.01%。 Using intelligent PID control, the peristaltic pump automatically replenishes acid / alkali; It can make the pH control performance very good, and with historical and trend curve for analysis;
Temperature control	Temperature control module : with jacket, intelligent PID control; Including electric heaters, circulating pumps, detection devices, valves, safety protection devices, controls module and pipelines, and adopts a fully enclosed pipeline design.

	The temperature is controlled by chilled water (normal temperature water is used to lower temperature after sterilization) and electric heating for exchange.
	Temperature control range: 15 ~ 50 °C; accuracy: ± 0.1 °C; Resolution: 0.1 °C
Feeding control	Perfect feeding system (acid, alkali, antifoam, feeding material), which can be fed regularly and quantitatively;
	Four peristaltic pumps can be selected in sections on the control interface,
	Automatic feeding and manual feeding can be switched at will, which is convenient and flexible in actual use.
Antifoam control	Electrode detection, peristaltic pump supplies defoamer.
	PID automatic control and alarm, sensitivity 100 ~ 100000Ω,
	Control mode: PLC full digital signal processing, digital setting control
Liquid measurement	Use Mettler weighing module, accuracy : 1/3000
Lid lifting	Only for <1000L vessel
	Using Taiwan Limtek electric linear lift, safe and convenient. Lift the lid 40cm or more.
Seed transfer	The seed transfer pipeline of the fermentation tank is located at the bottom of the tank body, and the material is AISI 316L.
	The seed transfer pipeline is equipped with a sanitary diaphragm valve to realize the online sterilization function without dead angle;
Pipeline	Pipeline: All pipelines are sanitary grade, 316L material, and the sanitary contact part Ra <0.4um, using automatic orbital welding technology, 99.999% purity argon is used as protective inert gas, and the pipeline and the whole system are pickled and passivated.
Valves	Valves: All valves and meters installed on the pipeline where is in direct contact with raw water, steam, and compressed air are hygienic grade. The blind pipe meets the 3D requirements and does not cause secondary pollution to the product;
Safety	The bursting disc is installed on the tank lid; The steam pipeline (tank side) is equipped with a safety valve to prevent overpressure.
Mark	The high-temperature and high-pressure parts are marked with colors and warning signs.
	The pipeline direction and purpose are marked;
	Each valve is marked with serial number and description .
UPS	When power is fault, the valve will close automatically to prevent any out of control condition.
	Equipped with UPS to ensure that the entire program section, data, ratings, etc. are stored safely for at least 5 minutes when the power supply fails.

2. Control system

BAILUNBIO S Control System: on-site controller, the next crew for the Siemens S7-200 series PLC control system, it is a mature, stable, universal type of PLC, to suit a variety of automation applications, especially in manufacturing process control Application. Its modular, easy to implement distributed configuration and easy to master and so on, Siemens PLC control center has become a biological reaction process control implementation is economical and advanced control system, the system uses the touch screen display, on-site direct operations (menu type), all English menu and interface; also with the host computer for data transmission, data sampling and display tank, the system view of data analysis, reading settings.

10-inch LCD touch-sensitive screen as the display screen to display complete content-rich, user-friendly screen is simple, easy to operate. And disturbing each other. The control system can adapt to different range of sensors and actuators the output of the input signal with the signal anti-jamming system.

Software: Siemens S7+ FORCE CONTORL

Data collection and control module: Siemens S200 PLC controller and data collection and control module

1. Control

① Manual control: You can set the percentage of open valve or a valve opening ② automatic control mode: You can select continuous or PID PID control the switch ③ sequence control: the control of all parameters can be pre-set at least 10 control section, ④ for automatic segmentation remotely control: PC control ⑤ acceptable correlation control: the dissolved oxygen can select speed, air flow rate, tank pressure, control of feeding, etc.; pH can choose to control with acid and alkali

2. Measurement function: can feeding, bubble enemy, such as measurement of pH

3. XY-axis to chart real progress can be any expansion and contraction of graphics display, to help the operator determine the parameters of the interaction between the effects of the fermentation process to quickly modify and adjust

4. Data Processing

1) to show that all the parameters of the trend curve analysis

2) to store, display, analyze all the historical curve parameters

3) can display all the parameters of the batch report

4) copy the data directly with the USB

5) data storage format is compatible with EXCEL, EXCEL platform can be to deal with

5. Controller self-protection function: to set the password, others can not modify the fermentation parameters; and the controller is not lost due to power settings for each parameter

6. Can the process control parameters, each parameter can have 10 block can be set

7. Time display of each parameter recording interval can be adjusted

8. Two parameters can be carried out with the screen settings and calibration

9. show the controller is really running against, the panel instructed the state process system, process hardware, and motion status indicator for understanding the various devices (such as peristaltic pumps, circulating pumps, heating, mixing, etc.) work status, Fault identification can be used as display

10. Each parameter of the transmission are independent modules, without disturbing each other, for easy maintenance

11. PH, DO and other parameters of control and associated controls can be adjusted

3. PC software for optional with extra USD\$8,000

BAILUNBIO-Ver8.0 is a data recording and processing as one of the fermentation of special packages, with the English operation interface, data storage and backup, curve analysis, report generation analysis, anomaly analysis, for the fermentation and production process control and data processing bring a lot of convenience.

Main features:

1. Can record the fermentation time, temperature, pH, dissolved oxygen, rotational speed, air flow, pressure, fill quantity, the amount of foam enemy, acid dosage
2. Can be sent under the controlled variable settings and control parameters, remote operation
3. Can be set sequence control
4. Can type off-line parameters, and records show
5. Trend curve, historical curve shows

can display 1-10 (any setting) curve;

- You can select a different start time and the time period (8,24,48,72,96 and 144 hours);
- You can set the upper and lower curves show, in order to improve display resolution;
- You can choose any of several different batches (1-10) curve also shows that for comparative analysis;
- and Y-axis parallel to the secant by the mouse to drag the reading, followed by curve data is activated, the fermentation time (X axis) and the same time the data curve with the secant movement changes;

6. Report Printing

- can show significant 1-12 (any set) data, and print output;
- You can select a different start time and the time and can choose different time intervals (1,5,15,30,60 minutes);
- You can choose any of several different batches (any set) data, and print output,
- To comparative analysis;

7. curve printing

- You can show the curve was printed by color printer;

EXCEL platform can also be carried out in data processing and printing, is very convenient.

Key components

ITEM	DESCRIPTION	BRAND	COUNTRY
Vessel	Main material 316L+304jacket	BAILUN BIO	China
Stir motor	SEW	SEW	Germany
Speed Regulator	ABB	ABB	Germany
Control system	German Siemens S7-1200 series PLC control system + embedded industrial control computer touch screen + control circuit board + electrical box and various electrical appliances. Including cabinets, control cores, signal acquisition, transmitters, power drives and other types of actuators	BAILUN BIO	China
relay		schneider	France
4 pcs peristaltic pump	BL-0~60rpm\min	Watson Marlow	England

PH electrode	Hamilton	Hamilton	Switzerland
PH cable	Hamilton	Hamilton	Switzerland
DO electrode	Hamilton	Hamilton	Switzerland
DO cable	Hamilton	Hamilton	Switzerland
Pressure electrode	BD	BD	Germany
filter	1element 3"	Sartorius	Germany
Mass flow meter	600L/min 1/2"	Votglin	Switzerland